

Keshav Bathla

Senior Software Engineer | AI & Platform Systems

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Technical Skills

AI & Agentic Systems: LLM Agents, LangChain, LangGraph, LlamaIndex, Model Context Protocol (MCP), Tool Calling, Claude, OpenAI API

RAG & Retrieval: RAG (Retrieval-Augmented Generation), Vector DBs (pgvector, Pinecone, Chroma), Embeddings, Hybrid Search, Evals & Guardrails

Languages & Backend: Python, Go, TypeScript, JavaScript, SQL, FastAPI, Flask, Django, Node.js, Gin, WebSockets, gRPC, Microservices

Data, Cloud & Infra: PostgreSQL, Redis, Elasticsearch, Kafka, SQS, Redis Streams, AWS, Docker, Kubernetes, Grafana, Prometheus, New Relic

Experience

Harness | *Senior Software Engineer*

Bengaluru, India | Dec 2025 – Present

- **CodePulse Developer Agent:** Built CodePulse, an internal coding assistant (Go Slack bot and CLI) that helps engineers navigate repositories, debug issues, and run dev tasks. Integrated Go/Java LSP for accurate code search and connected tools for Jira, Confluence, Grafana, Harness APIs, and Postgres.
- **Automated Plan-to-PR Engine:** Built an execution flow where the agent creates a code change plan from a Slack or Jira request, waits for developer approval, then runs a sandboxed Harness CI pipeline to test and open a PR. Customer support teams now debug and fix customer issues through Slack without needing on-call engineers.
- **Self-Serve Feature Rollouts:** Built Slack workflows that let Product Managers safely toggle feature flags, set up canary rollouts, and run experiments on their own, cutting rollout wait times from days to minutes.
- **Incident Triage Assistant:** Built an on-call triage bot that links PagerDuty alerts with recent deploys, Grafana dashboards, and traces to explain what broke, cutting new-engineer on-call ramp-up time by 80%.
- **Automated Security Patching:** Built a bot that monitors repositories for critical and high-severity CVEs, patches vulnerable dependencies, verifies builds in CI, and opens PRs, cutting manual patching effort by 95%.
- **MCP Infrastructure:** Built production Model Context Protocol (MCP) servers and CLI integrations so tools like Cursor and Claude Code can safely run pipelines, query code, and check observability metrics inside Harness.

CoinSwitch | *Senior Software Engineer (Trade & Payments)*

Bengaluru, India | Jan 2022 – Dec 2025

- **Market Making System:** Built an in-house market-making platform (4 Python microservices for pricing, hedging, and order routing) that delivered \$30M+ in liquidity at 99.99% uptime, significantly lowering dependence on third-party market makers.
- **Portfolio Analytics:** Built and launched a real-time portfolio analytics service for 20M users, using Redis caching and optimized database queries to keep response times under 400ms during market spikes.
- **Derivatives Trading Engine:** Built a crypto futures matching and trading service in Go that handled up to 50,000 orders/minute in high-volatility markets with sub-200ms p99 latency and 99.999% uptime.
- **High-Scale Microservices:** Scaled 10+ core microservices (Trade, Rates, Auth, Payments, Portfolio) to handle 1M concurrent users at peak with zero downtime, maintaining 99.999% platform availability.
- **Payment Gateway Framework:** Built a config-driven payment plugin in Go, cutting new gateway onboarding time to under 10 minutes while processing 1,000+ transactions/minute across multiple asset classes.
- **Real-Time Rates Streaming:** Built an event-driven WebSocket service streaming live prices and order-book depth for 1,000+ crypto pairs to 1M concurrent users without dropped connections.
- **Engineering Quality & CI:** Set up automated PR validation pipelines and testing standards across engineering teams, preventing outages caused by unvetted code changes.

Blinkit | *Software Engineer (intern → full-time)*

Gurugram, India | Jan 2021 – Dec 2021

- **Microservices Migration:** Migrated consumer Product Listing and Discovery services from legacy Python 2/Flask to Python 3/FastAPI, modernizing the architecture and adding async request processing.
- **Performance & Latency Scaling:** Reduced Product Listing API latency from 1,000ms to under 100ms as the user base grew to 20M+; added monitoring and alert dashboards in Grafana and New Relic.
- **Consumer Features:** Built the Instant Replacement feature for fresh produce (Fruits & Vegetables) on the customer app, resolved display bottlenecks, and helped improve checkout completion.

Oppia Foundation (Google Summer of Code) | *Student Developer*

Remote | Nov 2018 – Jun 2019

- **Google Summer of Code 2019:** Selected as a student developer for GSoC 2019; improved page rendering speed by 2 seconds across Oppia educational pages and merged 10+ pull requests.
- **Full-Stack Contributions:** Contributed to the AngularJS migration, backend optimizations in Python, and performed safe third-party library upgrades without regressions.

Education

Jaypee Institute of Information Technology | *B.Tech, Electronics and Communication Engineering* Noida, India | 2017 – 2021